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ORIGINAL RESEARCH

A Little Pepper-Upper? Systematic Review of Randomized Controlled Studies on Capsaicinoids, Capsinoids, and Exercise Performance

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Capsaicinoids and capsinoids are bioactive compounds mostly found in peppers. Although preclinical studies have reported that these compounds can improve exercise performance due to transient receptor potential vanilloid subtype 1 (TRPV1)-mediated thermogenesis, sympathetic modulation, and releasing calcium, it is still unclear how they affect exercise performance in humans as ergogenic supplements. Conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses reporting guide 2020, this systematic review examined the ergogenic effect of capsaicinoids and capsinoids on exercise performance in healthy adults. A total of 19 randomized placebo-controlled trials were included in the study. Studies were accessed by searching five databases (PubMed, Scopus, SPORTDiscus, Web of Science, and Cochrane Library). The quality of the studies was evaluated using the Cochrane risk-of-bias assessment tool. According to the study results, 10 studies examining the effect of capsaicinoid and capsinoid supplements on exercise performance reported positive effects. Also, the effect of capsaicinoids and capsinoids on exercise performance is more pronounced in resistance training. This difference, which varies according to the type of exercise, may be due to the correlation between capsaicin transient receptor potential vanilloid subtype 1 and insulin-like growth factor-1.

***Keywords**:* capsaicin, capsiate, dihydrocapsaicin, training

Capsaicinoids are compounds accountable for the appealing pungency of Capsicum (chili pepper) species. The main source of natural capsaicinoids is chili peppers, which consist of capsaicin, dihydrocapsaicin, nordihydrocapsaicin, homodihydrocapsaicin, and homocapsaicin (Luo et al., 2011). Capsaicin and dihydrocapsaicin are essential representatives of capsaicinoids (Naves et al., 2019). Capsaicinoids have demonstrated antioxidant, antitumoral, and anti-inflammatory properties in vitro and in vivo animal studies (Bogusz et al., 2018; Chapa-Oliver & Mejia-Teniente, 2016). Capsaicin (CAP), 8-methyl-N-vanillyl-6-nonenamide (C18H27NO3), is found mostly in chili peppers and other spicy foods. CAP is the major active component of red peppers, and about 70% of the burn of hot peppers is attributed to CAP (Iwai et al., 2003; Ludy et al., 2012). Capsinoids are CAP-like compounds found in a type of sweet red pepper and include capsiate, dihydrocapsiate, and nordihydrocapsiate (Whiting et al., 2012). Although capsinoids are much less pungent than CAP, there are studies reporting that both are effective at reducing body fat in mice and humans, as well as increasing energy expenditure and fat oxidation (Ludy et al., 2012; Rogers et al., 2018; Yokoyama et al., 2020).

Previous studies have indicated that the action of CAP is mediated by transient receptor potential vanilloid subtype 1 (TRPV1; Jordt & Julius, 2002; Kwon et al., 2021). Activation of TRPV1 accelerates blood flow by inducing nitric oxide synthesis, a potent vasodilator, increases sports performance by providing better oxygenation of muscle tissue, and improves exercise endurance and energy metabolism by upregulating peroxisome proliferator-activated receptor-γ coactivator-1α in animal models (Luo et al., 2012; Yang et al., 2010). CAP also induces the rise of sympathetic modulation and thermogenesis in human studies. Increased plasma catecholamines have been suggested as a potential mechanism for the ergogenic effects of caffeine ingestion, and CAP is known to stimulate adrenal catecholamine secretion. These processes support energy expenditure, lipolysis, and lipid oxidation (Shin & Moritani, 2007). In addition, CAP implements peripheral effects on mouse muscle cells by boosting the release of calcium by the endoplasmic reticulum, allowing force production (Lotteau et al., 2013). The most probably limiting factors for sustaining exercise performance are creatine phosphate availability, accumulation of inorganic phosphate, latency effects on calcium used by the sarcoplasmic reticulum, and central nervous system drive (Glaister, 2005). Athletes therefore use stimulants such as ephedrine or caffeine, supplements that increase or improve creatine phosphate stores, or buffering agents such as sodium bicarbonate to improve exercise performance (Glaister et al., 2008; McNaughton et al., 2008; Mujika et al., 2000). Because of all these potential effects, CAP has received attention among those interested in exercise and sport sciences (de Moura e Silva, Cholewa, Billaut, et al., 2021).

In vitro and in vivo animal studies reporting the favorable effects of capsaicinoids and capsinoids have recently led to clinical trials. Therefore, we aim to elucidate the effect of capsaicinoids and capsinoids on exercise performance markers so that we can explain how these promising results are reflected in human studies. This systematic review is one of the comprehensive studies discussing capsaicinoids and the effect of capsinoids on exercise performance. A systematic review (de Moura e Silva, Cholewa, Billaut, et al., 2021) with eight human studies on similar topics and a meta-analysis with 14 studies (Grgic et al., 2022) have been published previously. Our study is currently the most up-to-date and comprehensive systematic review, as it includes 19 studies. In addition,

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we tried to include both mechanistic and methodological details in the evaluation of the outputs.

Methods

This systematic review was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses statement guideline 2020 (Page et al., 2021) with the following research question: Does capsaicinoid or capsinoid supplementation improve exercise performance in healthy and physically active adults? The study protocol has been registered in the International Prospective Register of Systematic Reviews (PROSPERO) (registration number CRD42022371436).

Search Strategy

Two researchers independently performed a systematic search of the literature across the PubMed/MEDLINE, Scopus, SPORTDiscus, Cochrane Library, and Web of Science databases using the keywords "capsaicin" OR "8-Methyl-N-Vanillyl-6-Nonenamide" OR "capsiate" OR "dihydrocapsaicin" OR "capsaicinoid" OR "nordihydrocapsaicin" OR "homodihydrocapsaicin" OR "homocapsaicin" OR "capsinoid" AND "exercise" OR "acute exercise" OR "aerobic Exercise" OR "exercise training" OR "isometric exercise" OR "physical activity." To keep the articles included in the review up-to-date, articles published in the last 15 years were preferred. For this reason, articles published since 2009 are included. The language was restricted to English only.

Inclusion and Exclusion Criteria

The studies that met the following criteria were included in the study using the populations, interventions, comparison, outcome, and study design strategy. Populations, interventions, comparison, outcome, and study design strategy is presented in Table 1.

The following are the exclusion criteria used in this study:

- Individuals under the age of 18 years or over the age of 65 years.
- Studies with no full-text access.
- Studies written in languages other than English.
- Studies that do not meet populations, interventions, comparison, outcome, and study design strategy criteria.
- Animal studies, in vitro studies, cross-sectional studies, reviews, comments, and letters.

Data Extraction

Two researchers independently screened the results based on titles and abstracts to eliminate duplicate studies and evaluate them based on the inclusion/exclusion criteria. The results of database searches were combined, and duplicate studies were removed manually and using Mendeley Reference Manager. For further evaluation, two independent researchers extracted the following data from the included studies:

- Name of the author(s), year, and country of the study.
- Study design.
- Participants' characteristics (sex, exercise status, and age).
- Type of exercise.
- Dose and duration of supplementation protocols.
- Relevant parameters.

Quality Evaluation

The revised Cochrane risk-of-bias tool for randomized trials was used to assess the quality of the included studies (Sterne et al., 2019). The revised Cochrane risk-of-bias tool for randomized trials evaluates the risk of bias in five domains including: Domain 1: bias arising from the randomization process, Domain 2: bias due to deviations from intended interventions, Domain 3: bias due to missing outcome data, Domain 4: bias in measurement of the outcome, and Domain 5: bias in selection of the reported result. All studies were classified as "low risk," "some concerns," or "high risk." The methodological quality of the included studies was evaluated independently by two authors.

Results

Search Results

Preferred Reporting Items for Systematic Reviews and Meta-Analyses flow diagram (Figure 1) shows that 822 studies were searched from all databases; 360 of these studies were manually and automatically identified as duplicates by Mendeley Reference Manager software (v.2.80.1; Elsevier). The remaining 462 studies were filtered based on their titles and abstracts, with 324 being excluded due to unrelated topics. Following the completion of both phases of the search process, 19 studies were included for analysis in this review.

Supplementation Protocols

Most of the included studies used the capsule form of CAP or capsiate. However, two studies (Cross et al., 2020; Langan & Grosicki, 2020) provided chewable supplementation. In total, five studies administer capsiate supplementation (6 and 12 mg) before exercise (Costa et al., 2020; de Moura e Silva, Cholewa, Jager,

Table 1 PICO(S) Criteria

PICO(S) criteria	Defines
Populations (P)	Healthy and physically active adults (18–65 years)
Interventions (I)	All supplements containing capsaicin or capsaicinoid derivatives (including direct supplement or extracts and food if ingredient has been analyzed)
Comparison (C)	Inert or pure placebos
Outcome (O)	Primary outcome: Exercise performance test results (time to exhaustion, lifted weight, muscle strength and force, number of repetitions, and summed and peak torque) Secondary outcome: Heart rate, fatigue, RPE, blood lactate concentration, and oxygen consumption
Study design (S)	Randomized controlled trials

Note. RPE = rating of perceived exertion.

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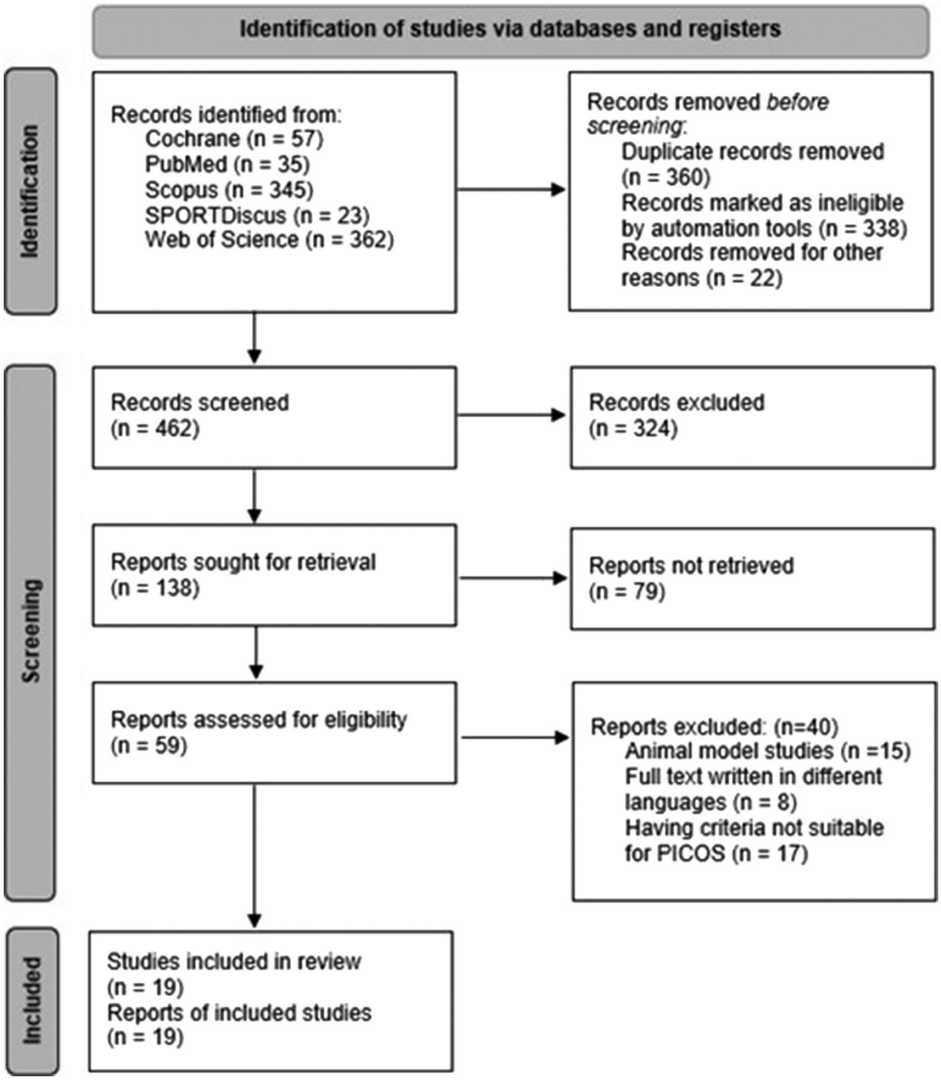

Figure 1 — PRISMA flow diagram. PRISMA = Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

et al., 2021; Dos Santos Gomes et al., 2022; Dos Santos Nunes de Moura et al., 2022; von Ah Morano et al., 2021). Josse et al. (2010) used capsinoid form (10 mg), and Walter et al. (2009) used capsicum extract form (33 and 34 mg). Most studies used starch as the placebo, but some also used dietary fiber (maltodextrin) or gelatin capsule (Opheim & Rankin, 2012; Volino-Souza et al., 2022). Supplements have been given to participants 45 min before exercise in most studies. But, it has been given participants supplementation 30 and 50 min before exercise in two studies (Giuriato et al., 2022; Walter et al., 2009).

Participants and Exercise Tasks

The studies included in this study were generally conducted with physically active adult males. However, three studies were conducted with active females and males (Cross et al., 2020; Langan & Grosicki 2020; von Ah Morano et al., 2021).

Exercise types are running (Costa et al., 2020, de Freitas, Cholewa, Gobbo, et al., 2018; de Freitas et al., 2019; Padilha et al., 2020; von Ah Morano et al., 2021), squat (de Freitas et al., 2022; de Freitas, Cholewa, Freire, et al., 2018; Simões et al., 2022), cycling (Giuriato et al., 2022, Josse et al., 2010, Langan & Grosicki 2020; Walter et al., 2009), leg and bench press (da Silva et al., 2022, de Moura e Silva, Cholewa, Jager, et al., 2021; dos Santos Nunes de Moura et al., 2022), knee and leg extension (Cross et al., 2020; dos Santos Gomes et al., 2022; Volino-Souza et al., 2022), and sprint (Opheim & Rankin, 2012).

Relevant Outcomes

The results of all studies included in the review are summarized in Table 2. From the studies, rating of perceived exertion (RPE), heart rate (HR), blood lactate concentration, and oxygen consumption (VO_2), which are exercise performance indicators, and exercise performance test results were obtained.

Rating of Perceived Exertion

A total of 14 of the included studies (Costa et al., 2020; de Freitas, Cholewa, Freire, et al., 2018; de Freitas, Cholewa, Gobbo, et al., 2018; de Freitas et al., 2019, 2022; Dos Santos Nunes de Moura et al., 2022; Giuriato et al., 2022; Josse et al., 2010; Langan & Grosicki, 2020; Opheim & Rankin, 2012; Padilha et al., 2020; Simões et al., 2022; von Ah Morano et al., 2021; Walter et al., 2009) evaluated RPE.

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Table 2 Summary of Included Studies

References	Country	Participants	Supplementation	Exercise task	Relevant outcomes	Notable significant ($p < .05$) findings
Costa et al. (2020)	Brazil	Twelve physically active men (age: 28.6 ± 5.4 years)	45 min before the intervention 12 mg capsiate or placebo	400- or 3,000-m time-trial running	RPE, HR, and exercise test results	• The time in seconds was significantly less in the capsiate compared to placebo groups (for the 3,000 m; $p = .015$, for the 400 m; $p = .046$). • No differences were found for RPE and HR between groups.
Cross et al. (2020)	United States	Nine young active females and males age: (age: 23.6 ± 1.5 and 24.2 ± 3.3 years, respectively)	45 min before the intervention 1.2 mg capsaicin or placebo	Isokinetic knee extensor contractile function test (per second, 120 maximal isokinetic knee extensions at 120° with passive flexion at 240°)	Exercise test results	• Peak torque was significantly greater in capsaicin than in the placebo ($p = .04$). • No significant differences were found for summed torque or fatigue index between the groups.
de Freitas et al. (2022)	Brazil	Eleven resistance-trained men (age: 23.3 ± 2.2 years)	45 min before the intervention 24 mg capsaicin or placebo	5-km HIIE, 10-min passive rest interval, and then 4 sets × 70% 1RM squat exercise (separated with 2-min rest interval)	RPE, HR, and exercise test results	• The number of repetitions ($p = .019$) and volume (Repetitions × Weight lifted [in kilograms]; $p = .028$) were significantly greater in the capsaicin compared with the placebo group. • RPE and mean HR were significantly lower in the capsaicin group compared with the placebo.
da Silva et al. (2022)	Brazil	Twenty male Brazilian Jiu-Jitsu athletes (age: 24.3 ± 1.5 years)	45 min before the intervention 12 mg capsaicin or placebo	Four sets of five repetitions of free bench press exercise at 60% of body mass (separated with 5-min rest interval)	Exercise test results	• The mean power output ($p = .001$) and mean peak velocity ($p < .001$) were significantly higher in capsaicin compared with the placebo during the exercise.
de Freitas, Cholewa, Freire, et al. (2018)	Brazil	Ten resistance-trained men (age: 22.7 ± 4.0 years)	45 min before the intervention 12 mg capsaicin or placebo	4 sets × 70% of the 1RM squat exercise (separated with 90-s rest interval)	RPE, lactate, and exercise test results	• The number of repetitions and total weight lifted (in kilograms; $p = .002$) were greater in the capsaicin group than placebo during the exercise. • RPE was significantly lower in the capsaicin group compared with the placebo group. • No difference was found for blood lactate concentration between groups.
de Freitas et al. (2019)	Brazil	Thirteen physically active men (age: 24.4 ± 4.0 years)	45 min before the intervention 12 mg capsaicin or placebo	HIIE running, for 15 s at 120% of VO_2max, interspersed by 15 s of passive recovery	RPE, VO_2, lactate, and exercise test results	• Time to exhaustion was longer in the capsaicin group than placebo ($p < .001$). • No differences were found for blood lactate concentration, RPE, and VO_2 between groups.
de Freitas, Cholewa, Gobbo, et al. (2018)	Brazil	Ten physically active men (age: 23.5 ± 1.9 years)	45 min before the intervention 12 mg capsaicin or placebo	1,500-m time-trial running	RPE, lactate, and exercise test results	• The time in seconds ($p = .009$) was significantly less in the capsaicin group compared with the placebo. • No difference was found for blood lactate concentration between groups. • RPE was significantly lower in the capsaicin group compared with the placebo.
de Moura e Silva, Cholewa, Jager, et al. (2021)	Brazil	Twenty untrained men (age: 18–30 years)	6 weeks 12 mg capsiate or placebo	6-week resistance training program with capsiate or placebo intervention, and then acute resistance exercise (leg and bench press)	Exercise test results	• Muscle strength at leg press did not differ significantly between the groups. • Capsiate significantly increased relative upper body strength in the capsiate group compared the placebo.

(continued)

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Table 2 *(continued)*

References	Country	Participants	Supplementation	Exercise task	Relevant outcomes	Notable significant ($p < .05$) findings
Dos Santos Nunes de Moura et al. (2022)	Brazil	Twenty resistance-trained men (age: 28 ± 4.1 years)	45 min before the intervention 6 mg capsiate, 12 mg capsiate, or placebo	4 sets × 70% of the 1RM bench press exercise (separated with 2-min rest interval)	RPE, lactate, and exercise test results	• Total weight lifted ($p = .039$) and the number of repetitions ($p = .043$) were significantly higher in the capsiate groups compared with the placebo. • RPE and blood lactate concentration did not differ significantly between the groups.
Dos Santos Gomes et al. (2022)	Brazil	Thirteen healthy young men (age: 25.2 ± 3.2 years)	45 min before the intervention 12 mg capsiate or placebo	5 × 10-s maximal voluntary isometric contraction test (knee extension; separated with 45-s rest interval)	Exercise test results	• Capsaicin supplementation improved peak force and fatigue index compared to placebo ($p < .05$). • The mean and minimum force did not change significantly between groups.
Giuriato et al. (2022)	United States	Ten healthy young men (age: 22.3 ± 3.6 years)	50 min before the intervention ~1.914 mg capsaicin + 0.658 mg dihydrocapsaicin or placebo	Time-to-exhaustion tests with constant-load cycling exercise (85% maximal work rate)	RPE, HR, VO_2, and exercise test results	• No significant differences were found for time to exhaustion, RPE, HR, and VO_2 between the groups.
Josse et al. (2010)	Canada	Twelve healthy active young men (age: 24.3 ± 3 years)	45 min before the intervention 10 mg capsinoid or placebo	Cycling at 55% VO_2max, and for 30 min into recovery	RPE, HR, VO_2, and lactate	• Resting VO_2 was higher in capsinoid group. • RPE, HR, and blood lactate concentration did not change significantly between groups.
Langan and Grosicki (2020)	United States	Thirteen active males and females (age: 25.3 ± 3.1 and 22.3 ± 2.1 years, respectively)	45 min before the intervention 1.2 mg capsaicin or placebo	Time-to-exhaustion tests on a cycle ergometer at a workload eliciting ~90% VO_2max	RPE, HR, and exercise test results	• No significant differences were found for time to exhaustion, RPE, and HR between the groups.
Opheim and Rankin (2012)	United States	Nineteen healthy male experienced athletes or runners (age: 22.6 ± 2.6 years)	45 min before the intervention 25.8 mg capsaicin or placebo	Repeated sprint test consisting of 15- to 30-m maximal effort sprint (separated with 35-s rest interval)	RPE and exercise test results	• Mean repeated sprint test times and RPE did not differ significantly between groups.
Padilha et al. (2020)	Brazil	Fifteen trained men (age: 22.1 ± 3.07 years)	45 min before the intervention 12 mg capsaicin or placebo	Running at 90% of VO_2max to exhaustion	RP, HR, lactate, and exercise test results	• RPE, HR, blood lactate concentration, and time to reach $\dot{V}O_{2\,max}$ did not differ significantly between groups.
Simões et al. (2022)	Brazil	Eleven young resistance-trained men (age: 21.5 ± 2.1 years)	45 min before the intervention 12 mg capsaicin, 400 mg caffeine, 12 mg capsaicin + 400 mg caffeine, or placebo	4 sets × 70% 1RM squat exercise (separated with 90-s rest interval)	RPE, HR, and exercise test results	• The total volume, RPE, and HR between the four experimental conditions did not change.
Volino-Souza et al. (2022)	Brazil	Eight active males (age: 33.33 ± 7.42 years)	45 min before the intervention 12 mg capsaicin or placebo	Three sets of leg extension exercise at 70% of 1RM until muscle fatigue	Lactate and exercise test results	• Blood lactate concentration, number of repetitions, and time of exercise did not differ significantly between groups.
von Ah Morano et al. (2021)	Brazil	Twenty-one male amateur athletes (age: 29.3 ± 5.5 years)	45 min before the intervention 24 mg capsiate or placebo	10-km time-trial running	RPE, HR, lactate, and exercise test results	• Blood lactate concentration, RPE, HR, and running time did not differ significantly between groups.
Walter et al. (2009)	United States	Twenty healthy men (age: 21.5 ± 1.4 years)	30 min before the intervention 200 mg caffeine, 33.34 mg capsicum extract, 20 mg niacin, and 5 mg black pepper extract or placebo	Cycling at 80% of VO_2max to exhaustion followed by 1RM leg and bench press test	RPE and exercise test results	• RPE and muscular strength did not differ significantly between groups.

Note. RPE = rating of perceived exertion; HIIE = high-intensity intermittent exercise; 1RM = one-repetition-maximum; VO_2max = maximal oxygen consumption; HR = heart rate.

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In total, three studies (de Freitas, Cholewa, Freire, et al., 2018; de Freitas, Cholewa, Gobbo, et al., 2018; de Freitas et al., 2022) reported that the RPE was reduced in the intervention groups.

Heart Rate

In total, eight of the included studies (Costa et al., 2020; de Freitas et al., 2019; Giuriato et al., 2022; Josse et al., 2010; Langan & Grosicki, 2020; Padilha et al., 2020; Simões et al., 2022; von Ah Morano et al., 2021) evaluated HR. One study (de Freitas et al., 2022) reported that the HR was reduced in the intervention groups.

Blood Lactate Concentration

A total of eight of the included studies (de Freitas, Cholewa, Freire, et al., 2018; de Freitas, Cholewa, Gobbo, et al., 2018; de Freitas et al., 2019; Dos Santos Nunes de Moura et al., 2022; Josse et al., 2010; Padilha et al., 2020; Volino-Souza et al., 2022; von Ah Morano et al., 2021) evaluated blood lactate levels. None of the studies reported significant between-group changes in blood lactate concentrations.

Oxygen Consumption

Three of the included studies' (de Freitas et al., 2019; Giuriato et al., 2022; Josse et al., 2010) outputs related to VO_2 and muscle oxygenation during or after exercise. Josse et al. (2010) reported that resting VO_2 was significantly higher in intervention group compared the placebo group.

Performance Test

In total, 10 of the studies (Costa et al., 2020; de Freitas, Cholewa, Gobbo, et al., 2018; de Freitas et al., 2019; Giuriato et al., 2022; Langan & Grosicki, 2020; Opheim & Rankin, 2012; Padilha et al., 2020; Volino-Souza et al., 2022; von Ah Morano et al., 2021; Walter et al., 2009) evaluated performance as time-dependent. A total of five studies (de Freitas, Cholewa, Gobbo, et al., 2018; de Freitas et al., 2022; de Moura e Silva, Cholewa, Jager, et al., 2021; Dos Santos Nunes de Moura et al., 2022; Volino-Souza et al., 2022) reported that measuring the number of repetitions. A total of seven studies (Cross et al., 2020; da Silva et al., 2022; de Freitas, Cholewa, Freire, et al., 2018; de Moura e Silva, Cholewa, Jager, et al., 2021; Dos Santos Gomes et al., 2022; Dos Santos Nunes de Moura et al., 2022; Simões et al., 2022) evaluated muscle strength, force, lifted mass weight, torque, or total volume (Lifted Weight × Number of Repetitions). According to the performance test results, 10 studies (Costa et al., 2020; da Silva et al., 2022; de Freitas, Cholewa, Freire, et al., 2018; de Freitas, Cholewa, Gobbo, et al., 2018; de Freitas et al., 2019, 2022; de Moura e Silva, Cholewa, Jager, et al., 2021; Dos Santos Gomes et al., 2022; Dos Santos Nunes de Moura et al., 2022) reported that exercise performance increased significantly in the intervention group.

Quality Evaluation

For all domains (1, 2, 3, 4, and 5), the classification for the included studies was scored "low risk" or "some concerns." The overall risk-of-bias assessment on the revised Cochrane risk-of-bias tool for randomized trials for all included studies was "some concerns." Figure 2 displays the evaluation of the risk of bias for all included trials.

Discussion

Exercise performance test results, which was our primary output in our study, 10 of 19 studies reported positive effects on improved performance in the intervention group. While seven of these 10 studies that reported the advantageous effects of capsaicinoids and capsinoids supplements tested the exercise performance with resistance training (Cross et al., 2020; da Silva et al., 2022; de Freitas, Cholewa, Freire, et al., 2018; de Freitas et al., 2022; de Moura e Silva, Cholewa, Jager, et al., 2021; Dos Santos Gomes et al., 2022; Dos Santos Nunes de Moura et al., 2022). Mechanistically, what we know so far gives us some ideas that capsaicinoids and capsinoids will improve exercise performance. As we mentioned previously, the most focus is on TRPV1 activation. CAP induces TRPV1 activation, which modulates muscle contraction by releasing Ca^{+2} ions into the cell or increasing sympathetic activity by inducing catecholamines (Iida et al., 2003; Kalmar, 2005; Lotteau et al., 2013). This improves muscle actin–myosin interaction, force generation processes, and intramuscular triglyceride consumption. This results in preserving muscle glycogen and increased time to exhaustion (Lotteau et al., 2013). Another mechanism is that TRPV1 partially increases fat metabolism through its neuroadrenergic effects, which may positively affect performance in long-term sports (Oshioka et al., 1995).

However, these possible physiological mechanisms cannot fully explain why CAP is more effective in resistance exercise than in endurance exercise. Comparing it with the result of the study, which reported a positive effect of CAP and capsiates on resistance exercises, previously published by Grgic et al. (2022), it seems to be compatible with our findings. In addition to those studies, we reported two more resistance exercise studies reporting significant results. But we also hope to explain the reason for this difference here. One of the main physiological differences between endurance and resistance exercises is that endurance exercises are mostly dependent on mitochondrial biogenesis, whereas resistance exercises are mostly dependent on ribosomal biogenesis. This indicates that endurance exercise performance mostly depends on adenosine monophosphate-activated protein kinase (AMPK) activity, while resistance exercises depend on mammalian target of rapamycin complex 1 (mTORC1) activity. AMPK activity is dependent on energy sources (glucose, triglycerides), while mTOR activation requires positive signals from both nutrients and growth factors. Insulin-like growth factor-1 (IGF-1) induces phosphoinositide 3 kinase/AKT serine/threonine kinase 1/mTORC1 pathway, protein synthesis, and muscle hypertrophy (Li et al., 2014; Mesquita et al., 2021). It has been reported that there is a link between the insulin family and vanilloid receptors (Li et al., 2014; Rosta et al., 2022). Lilja et al. (2007) have previously shown that IGF-1 and insulin upregulate TRPV1 protein levels. Van Buren et al. (2005) also reported that TRPV1 is sensitized and translocated by insulin and IGF-1. In this case, increased IGF-1 signaling in acute resistance exercise may help to explain the positive effects by modulating TRPV1 (Figure 3).

Based on the results of the quality assessment, four studies (Cross et al., 2020; Giuriato et al., 2022; Langan & Grosicki, 2020; Opheim & Rankin, 2012) used a single-blind design and were methodological of lower quality. If even one of the domains included in the quality evaluation is evaluated as having some concerns, the resulting quality of the study is evaluated in that way. Although all of the studies included in the systematic review were randomized controlled studies, all of them were concluded as having some concerns because there was no explanation about how the randomization was made.

Secondary outcomes of our study focus on RPE, fatigue, HR, blood lactate concentration, and VO_2. The RPE assesses an individual's physical activity intensity based on physical sensations such as breathing, HR, sweating, and muscle fatigue

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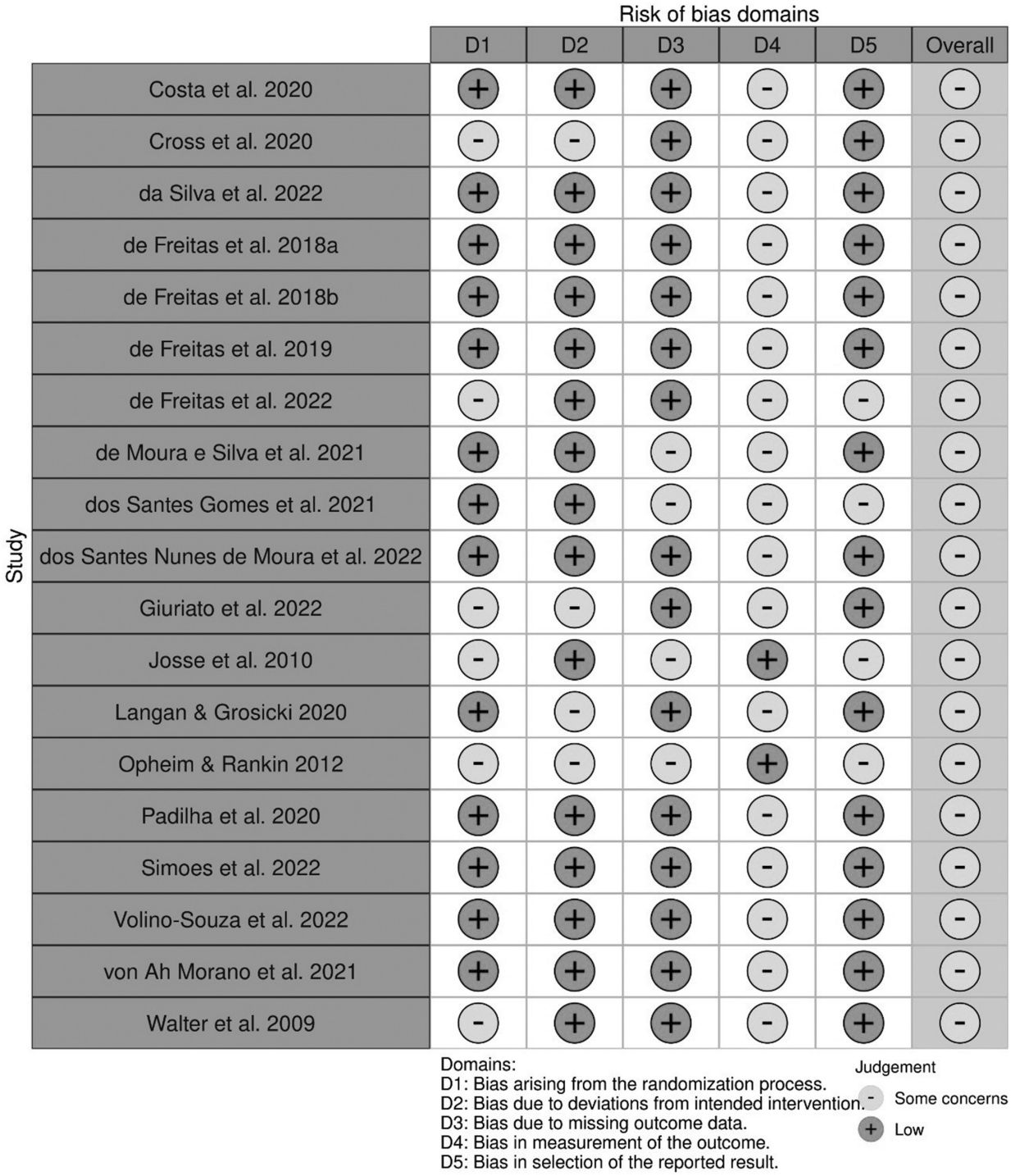

Figure 2 — Quality evaluation of included studies.

experienced by the individual during physical work. The studies we included used three different RPE scales (Borg, 1998; Robertson, 2004). This indicates that the RPE results of the studies we included have a heterogeneous distribution and confounding in terms of meta-analysis. Although the scales seem easy to apply and have found a wide range of applications, there are various situations that should be considered methodologically (Lopes et al., 2022). First , since the scales are in English, the realization of language and culture adaptation will affect the results (Cabral et al., 2020). Ideally, it should be applied during or immediately after exercise. In addition, individuals should be encouraged to distinguish somatic sensations, avoid judgments about honesty, and exercise intensity (Lopes et al., 2022). Among the studies in which the scale was used in nonnative English countries, Padilha et al. (2020) was the only study that used the Portuguese validated version and included methodological details.

There is a high correlation between RPE and HR (Williams, 2017). When evaluating RPE, an increased workload for the same effort is associated with increased performance. Less HR can be evaluated similarly in response to more workload (Pageaux, 2016).

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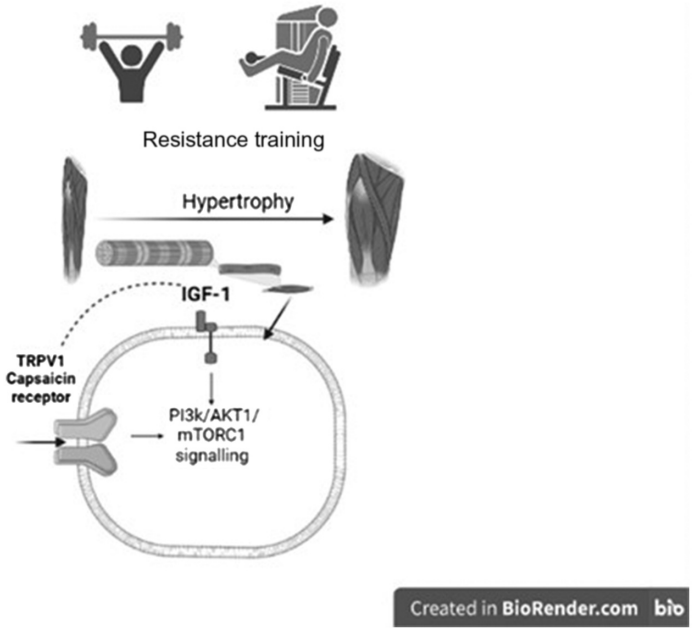

Figure 3 — Relationship between TRPV1 and IGF-1 in resistance exercise. TRPV1 = transient receptor potential vanilloid subtype 1; IGF-1 = insulin-like growth factor-1.

However, accumulated fatigue in interval training can result in increased effort despite decreased HR (Buchheit et al., 2012). Therefore, monitoring HR and RPE together may help in better evaluation (Lopes et al., 2022). HR responses of our included studies seem to be compatible with RPE.

Another condition that determined the performance of exercise is skeletal muscle fatigue. Muscle fatigue can be monitored and evaluated using a variety of methods (Finsterer & Mahjoub, 2014; Roete et al., 2021). The lactic acid hypothesis, which is defined as the accumulation of lactate and acidosis in the working muscle, causing the inhibition of contraction in the muscles, and the decrease in exercise performance, has become controversial for the last decades, and it is thought that it is not appropriate to use it alone as an indicator of fatigue (Theofilidis et al., 2018). Even, lactate has been shown to support aerobic adaptations by regulating intramuscular triglyceride metabolism (Cairns, 2006). Blood lactate measurements are used as a practical method of estimating the acid–base status in muscle. This assumes that blood lactate reflects muscle lactate, but this is not always the case, especially when lactate flow from working muscle and distribution of lactate to other muscles or tissues is uneven during intense and intermittent exercise. Other factors such as a previous exercise that reduces glycogen levels or a high-carbohydrate diet may affect blood lactate concentration and the timing of reaching predefined thresholds (Jamnick et al., 2018; Juel, 2008). Therefore, it needs appropriate to follow diet and exercise protocols for accurate evaluation. None of the studies reported significant changes during or after exercise lactate levels between the groups. However, only von Ah Morano et al. (2021) evaluated the difference between macronutrient distribution 24 hr before each trial group.

The cardiopulmonary exercise test, which is another exercise performance indicator, provides a complete assessment of the physiological responses of the pulmonary, cardiovascular, muscle, and cellular oxidative systems to exercise (Lima-Silva et al., 2013). Maximum VO_2 is the most important parameter for evaluating cardiopulmonary capacity, especially in endurance sports, and it is the human body's maximum oxygen uptake and defines the maximum amount of energy that can be accessed by aerobic metabolism at the peak of exercise (Guazzi et al., 2017; Zinner et al., 2015). Athletes in different sports disciplines have a wide range of maximum VO_2, so for better interindividual comparisons, it is better to express this as a percent-predicted value or in milliliters of oxygen per kilogram of body weight per minute. In fact, it is recommended to use lean mass for interdisciplinary comparisons, since the ideal body weight varies (Mazaheri et al., 2021; Mezzani, 2017). A total of three of the included studies (de Freitas et al., 2019; Giuriato et al., 2022; Josse et al., 2010) reported results related to VO_2. Josse et al. (2010) reported that resting VO_2 levels were higher in the intervention group.

Limitations

We have generally adapted to our preregistration plan. We had planned to include pepper consumption outside of supplements. However, since the phytochemical amount and bioavailability of pepper ingredients are changed, we considered it appropriate to include only supplements. Our initial intention was to conduct a meta-analysis on this topic, but we opted for a systematic review instead for the first phase of our research. However, we are not giving up on our original goal. We will keep track of the latest studies and incorporate them into our analysis. Our aim is to publish a meta-analysis study as the second phase of our research in the near future. We think that subgroup analyses are important. However, the current table is not yet sufficient to illuminate issues such as gender, dose, and type of exercise. In addition to revealing more studies worth reporting, our study also contains remarkable insight in terms of mechanism. We also drew attention to the methodological deficiencies in the studies as much as possible. Therefore, we think that it would be good to publish an updated meta-analysis in the coming years when good-quality studies are added to the literature. We believe that a systematic review would provide a more comprehensive and critical overview of the existing literature, as well as identify the gaps and limitations that need to be addressed in future studies.

Conclusions

Capsaicinoid and capsinoid supplementation seem to have positive effects, especially on resistance exercise performance. However, the reasons why this effect does not occur in endurance exercises should be investigated better. In addition, it is necessary to determine the appropriate dose and the side effects better. Many studies have not reported the side effects. Also, the effects of long-term use are not yet fully known.

Acknowledgments

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